

Pandas Assignment Problems

Q1. IRIS Dataset

Dataset: Iris Flower Dataset

A. Display the following information:

- First 10 rows
- Shape & Data types
- Summary statistics (mean, std, min, max)

B. Select only those rows where:

- `petal_length > 4.5`
- `species = "Iris-virginica"`

C. Group by **species** and compute:

- Average `sepal_length`
- Maximum `petal_width`
- Standard deviation of `sepal_width`

D. Create a new column:

`petal_ratio = petal_length / petal_width`

- For each species, find the average `petal_ratio`.

Q2. TITANIC Dataset

Dataset: Titanic Dataset

A. Display only the columns:

Name, Sex, Age, Fare, Survived

B. Select passengers who:

- Are female
- Fare > 30

C. Group by **Pclass** and compute:

- Survival rate
- Average Fare
- Average Age